RESPIRATORY EPITHELIUM

AND

HISTOLOGY OF TRACHEA

RESPIRATORY SYSTEM

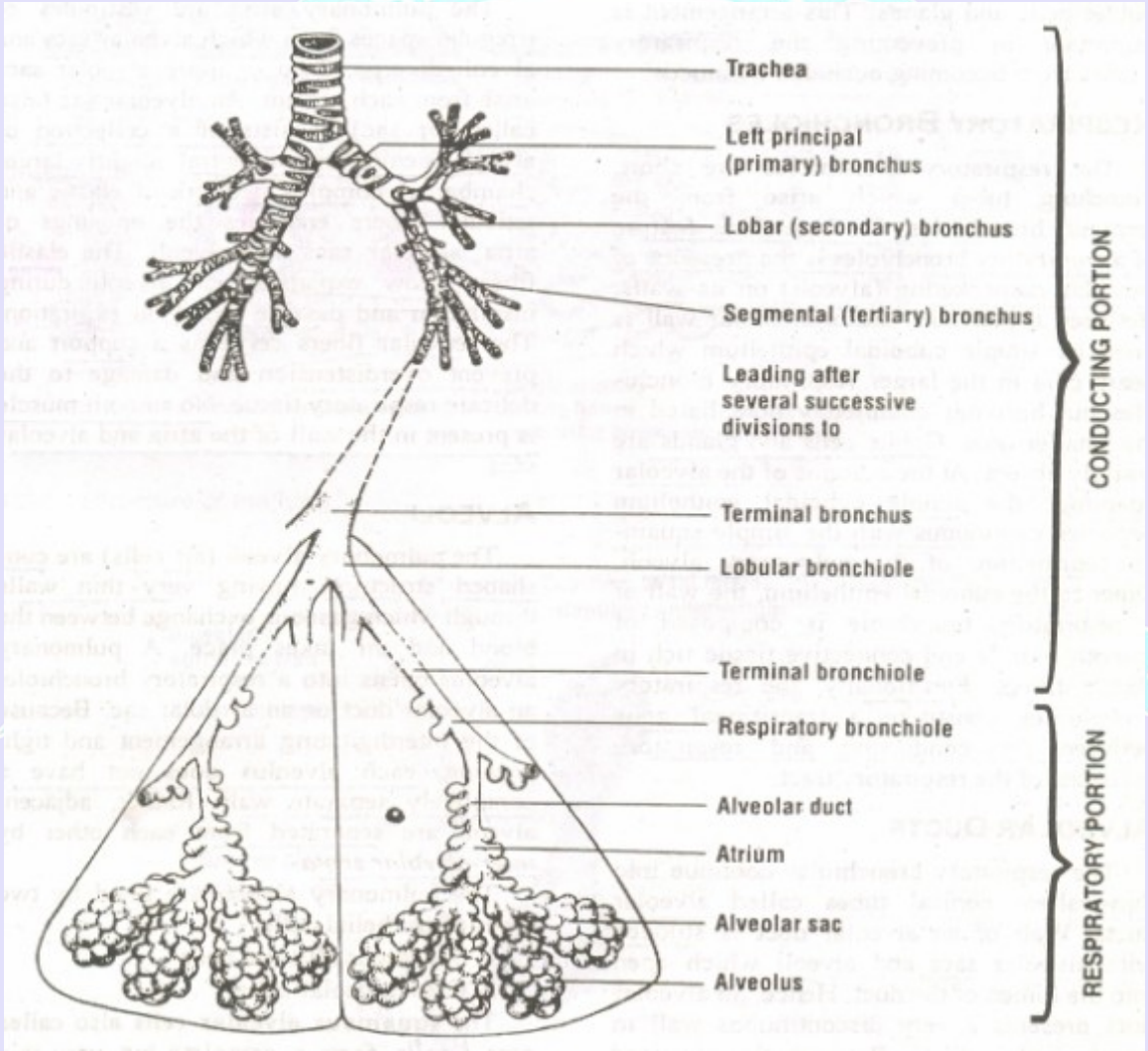
Consists of Lungs and Respiratory Passages.

- **Conducting portion**
  - **Nasal cavities**
  - **Pharynx**
  - **Larynx**
  - **Trachea**
  - **Bronchi**
  - **Bronchioles**
  - **Terminal bronchioles**
- **Respiratory portion**
  - **Respiratory bronchioles**
  - **Alveolar ducts**
  - **Alveolar sacs**
  - **Atria**
  - **Alveoli**

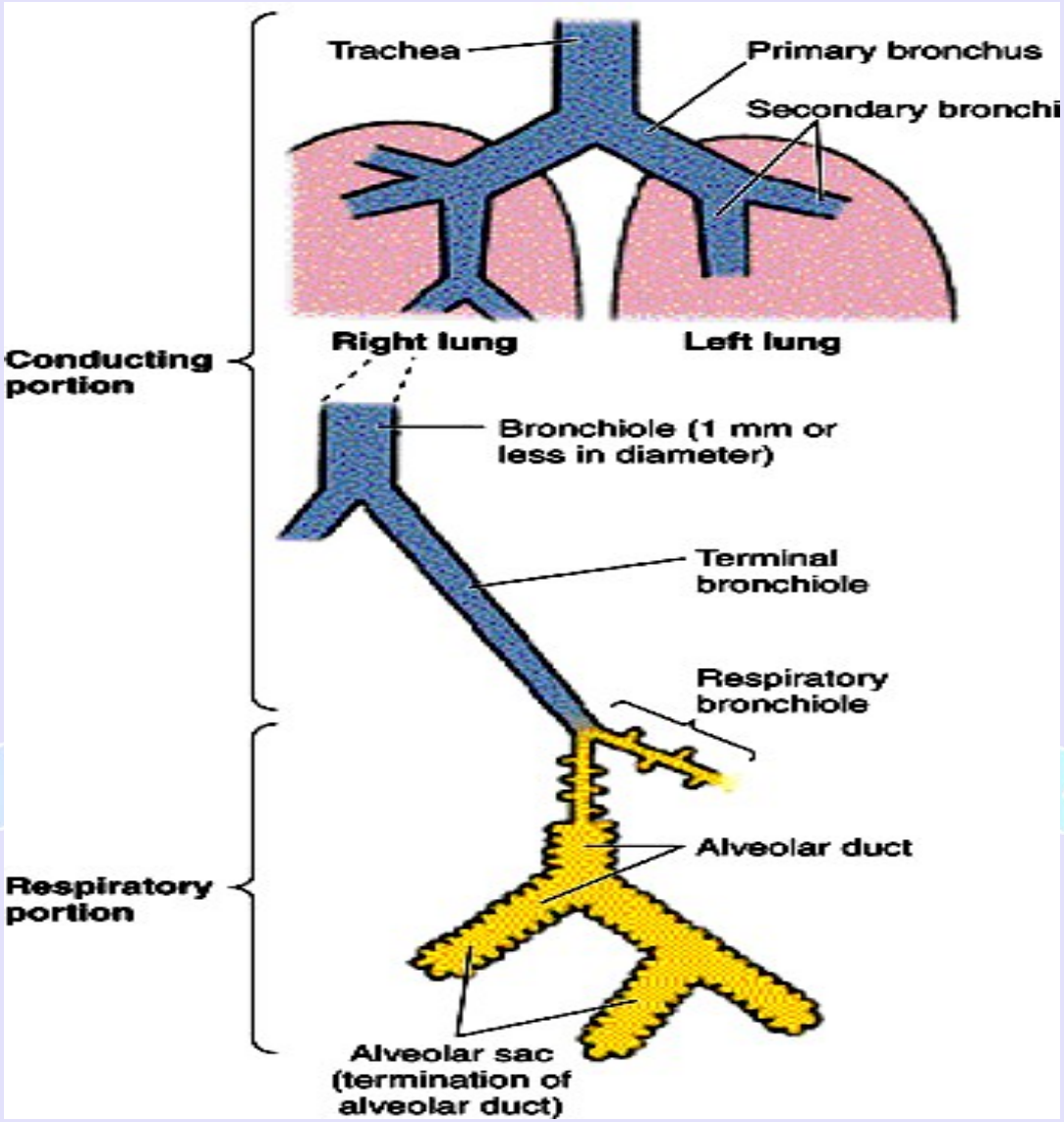
FUNCTIONS OF RESPIRATORY EPITHELIUM

- Exchange of O_2 and CO_2 between the blood and inhaled air across the alveoli
- Olfaction
- Phonation

CONDUCTING PORTION



RESPIRATORY PORTION



RESPIRATORY EPITHELIUM

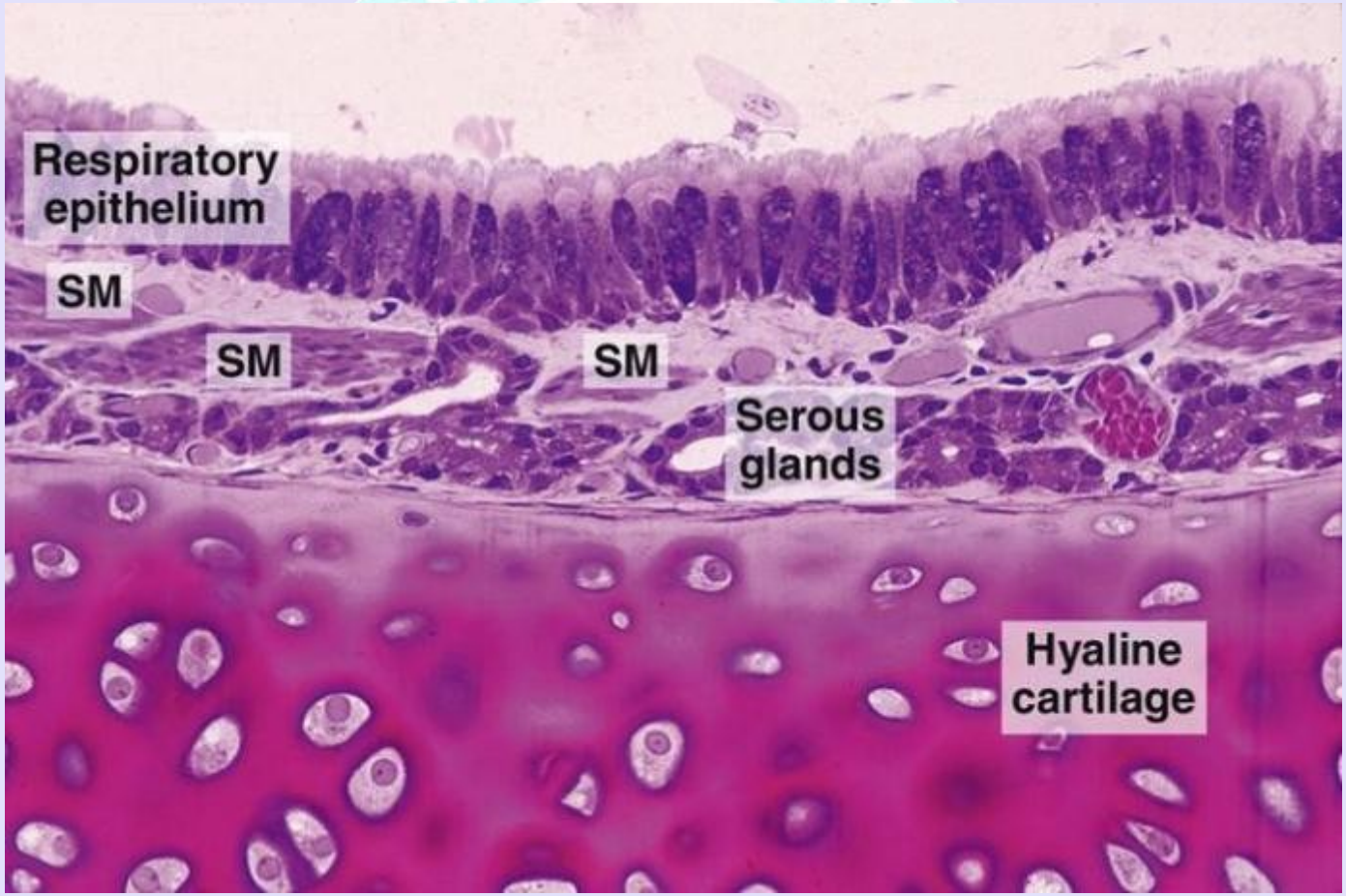
Following types of epithelia line the respiratory system:

- **Ciliated Pseudo stratified Columnar** epithelium-conducting portion up to the large bronchioles.
- **Ciliated Simple cuboidal** epithelium-terminal bronchioles
- **Simple squamous** epithelium-alveoli.

BRONCHIAL WALL CONSTITUENTS

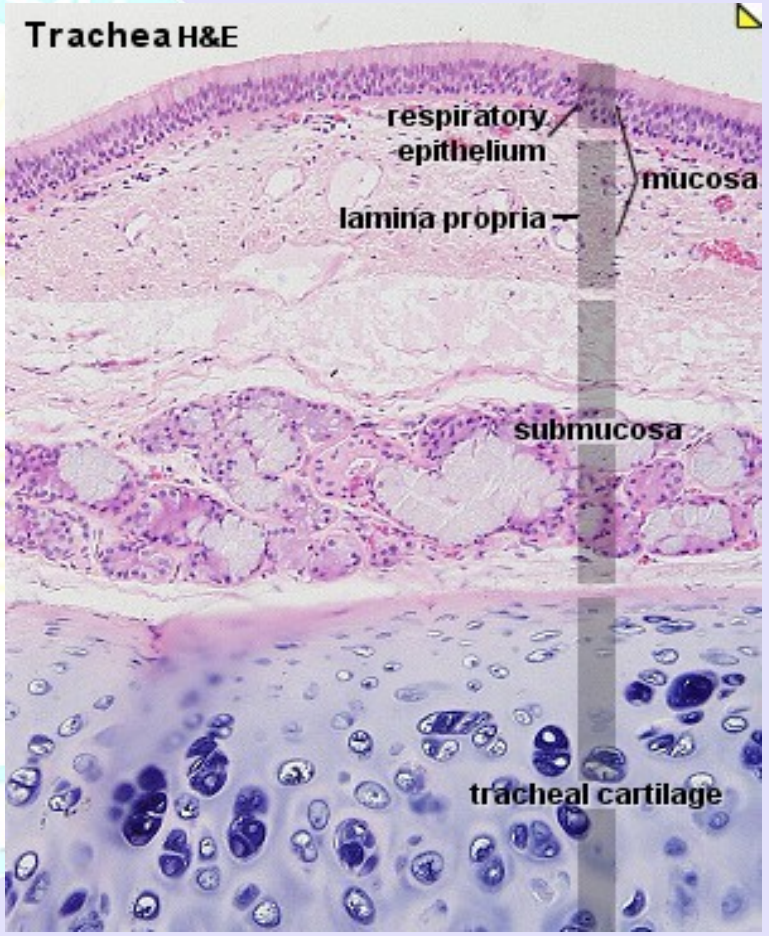
- Cilia
- Goblet cells
- Glands
- Cartilage
- Smooth muscle fibers
- Elastic fibers

SECTION THROUGH BRONCHIAL WALL SHOWING CONSTITUENT STRUCTURES.



CARTILAGE

- Found in larynx, trachea up to the smaller bronchi
- Trachea is stabilized by 16-20 C-shaped cartilages
- Small cartilage plates support the bronchi
- Cartilage is completely absent from bronchioles



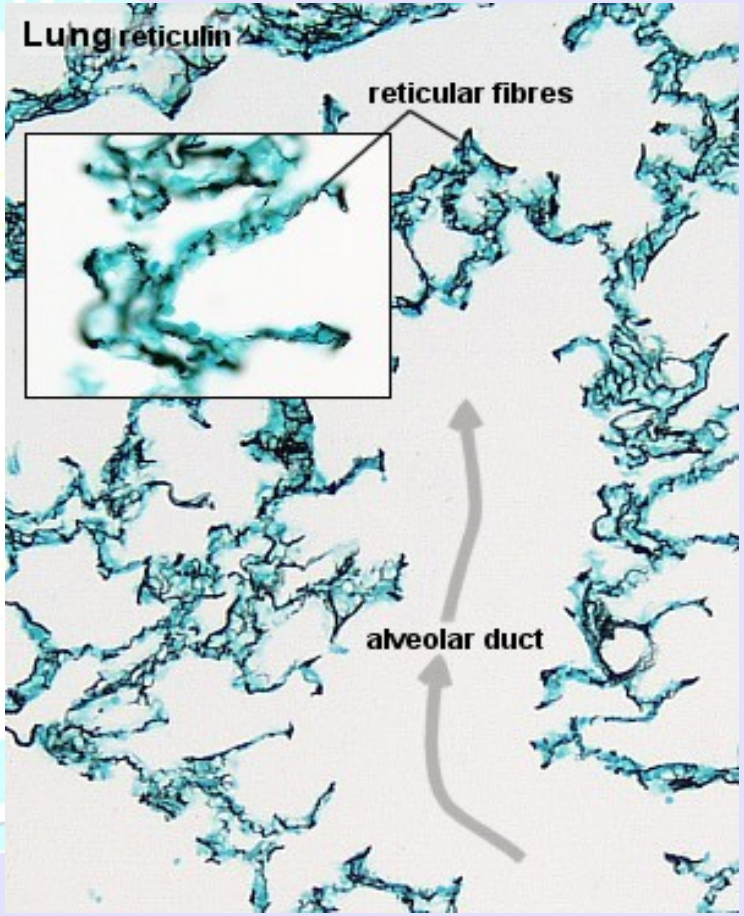
SMOOTH MUSCLE FIBERS

- Appear in the trachea
- Gradually thicken down the respiratory tree up to the bronchioles
- Form crisscross spiral bundles.



ELASTIC FIBERS

- Appear in nasopharynx
- Gradually increase in number
- Become abundant in respiratory portion



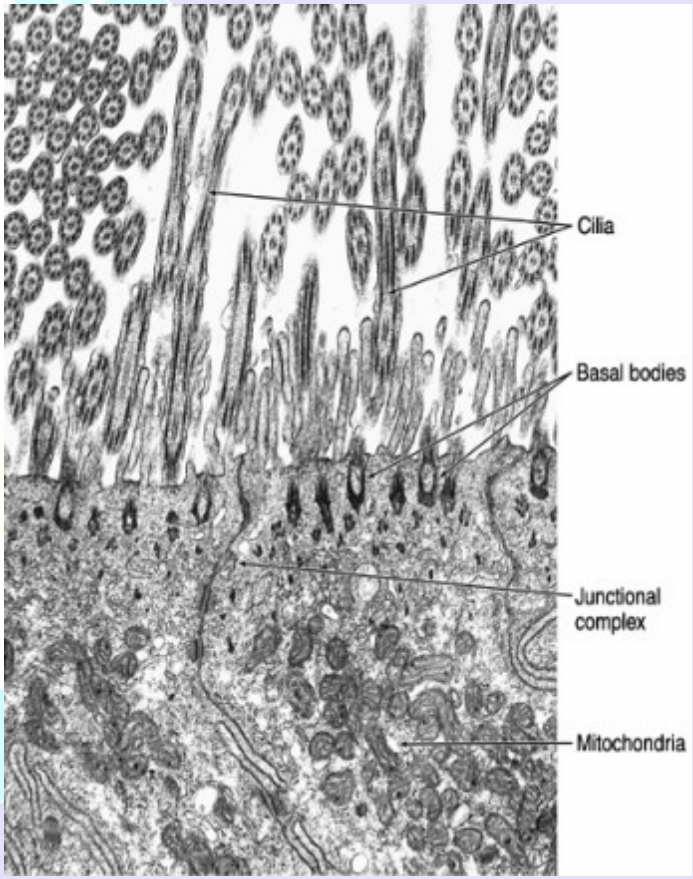
CELL TYPES

Electron microscopy reveals Six types of cells present in epithelia lining the **conducting portion**.

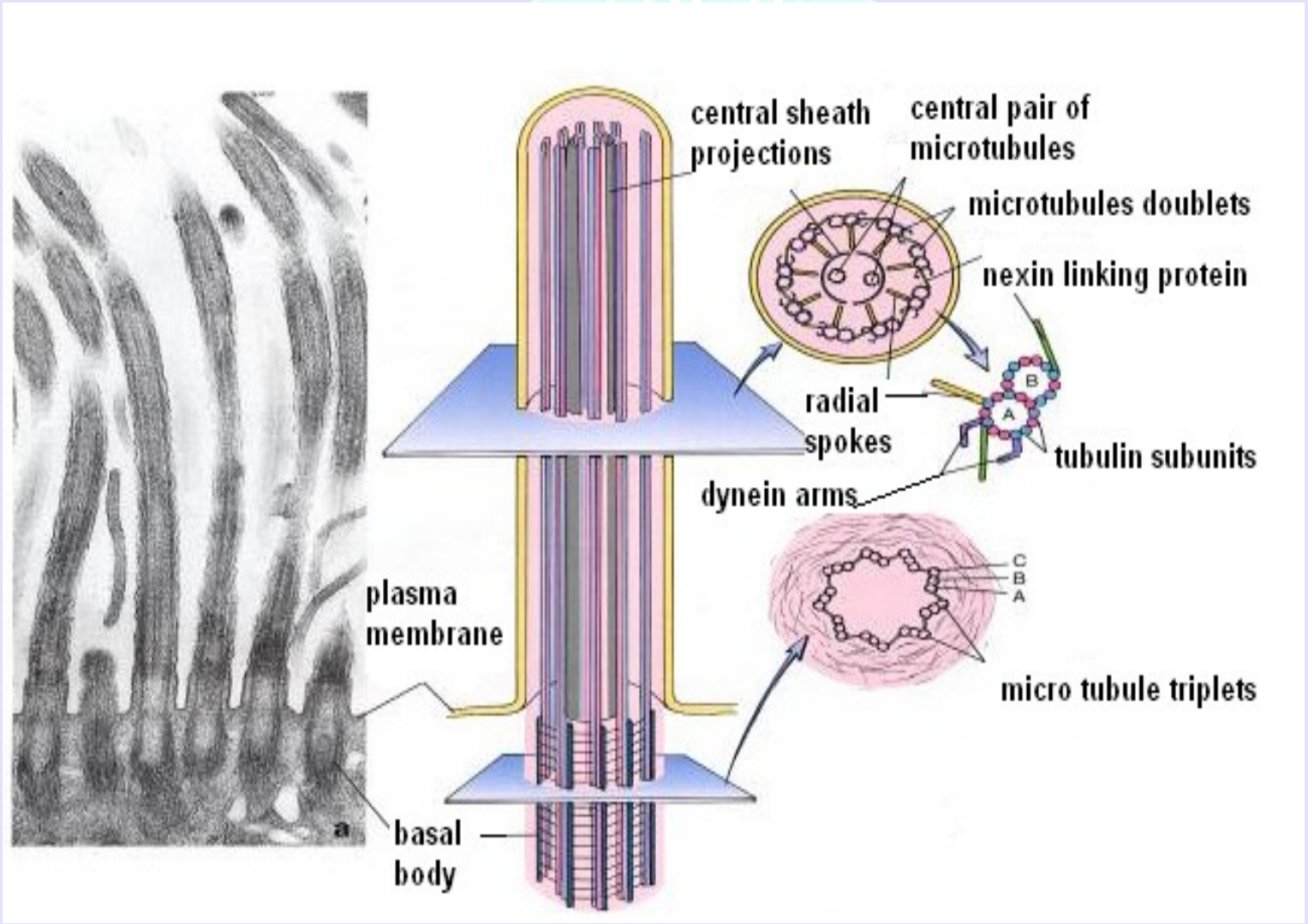
- **Ciliated columnar cells**
- **Goblet cells**
- **Brush cells**
- **Basal (short) cells**
- **Small granular cells**
- **Clara cells**

CILIATED COLUMNAR CELLS

- Most abundant cell type
- Each cell has up to 300 ciliary tubules on the apical surface.
- Beneath the apical surface of the cell there are basal bodies and numerous mitochondria.
- Basal bodies serve as the anchoring sites for ciliary axonemes

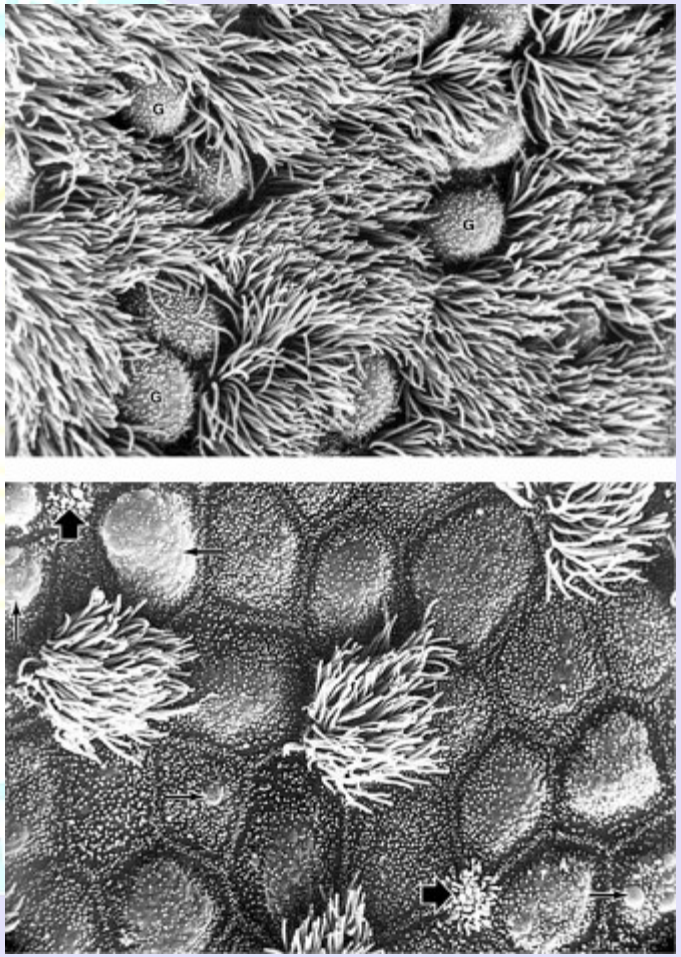


STRUCTURE OF A CILIARY MICROTUBULE



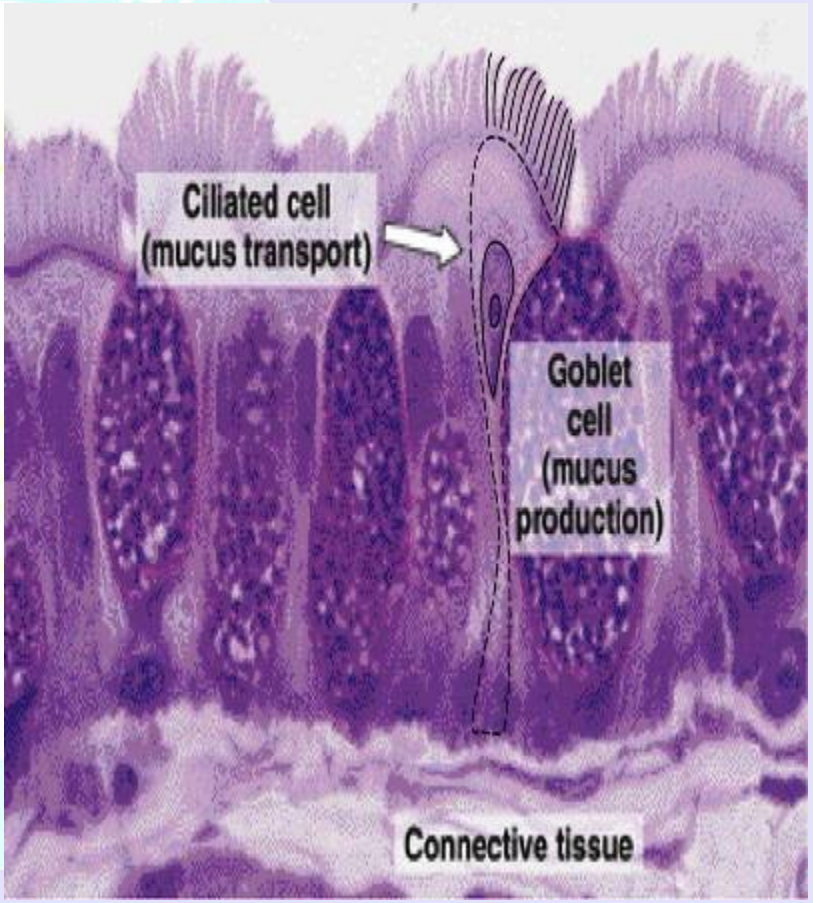
BRUSH CELLS

- Columnar cells having numerous microvilli on apical surface
- Afferent nerve endings innervate them on their basal surface
- Considered Sensory Receptors.
- Brush cells are indicated by thick arrows in the bottom part of EM photograph.

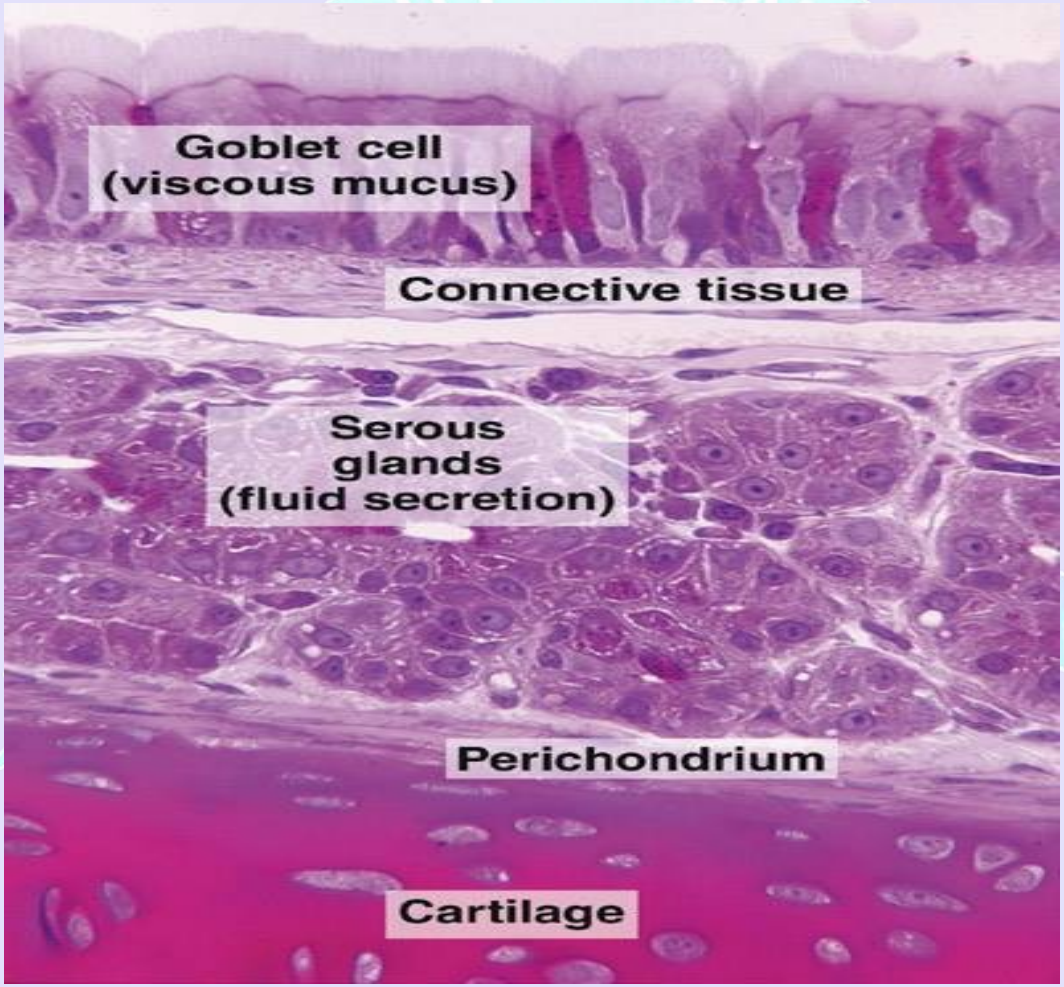


GOBLET CELLS

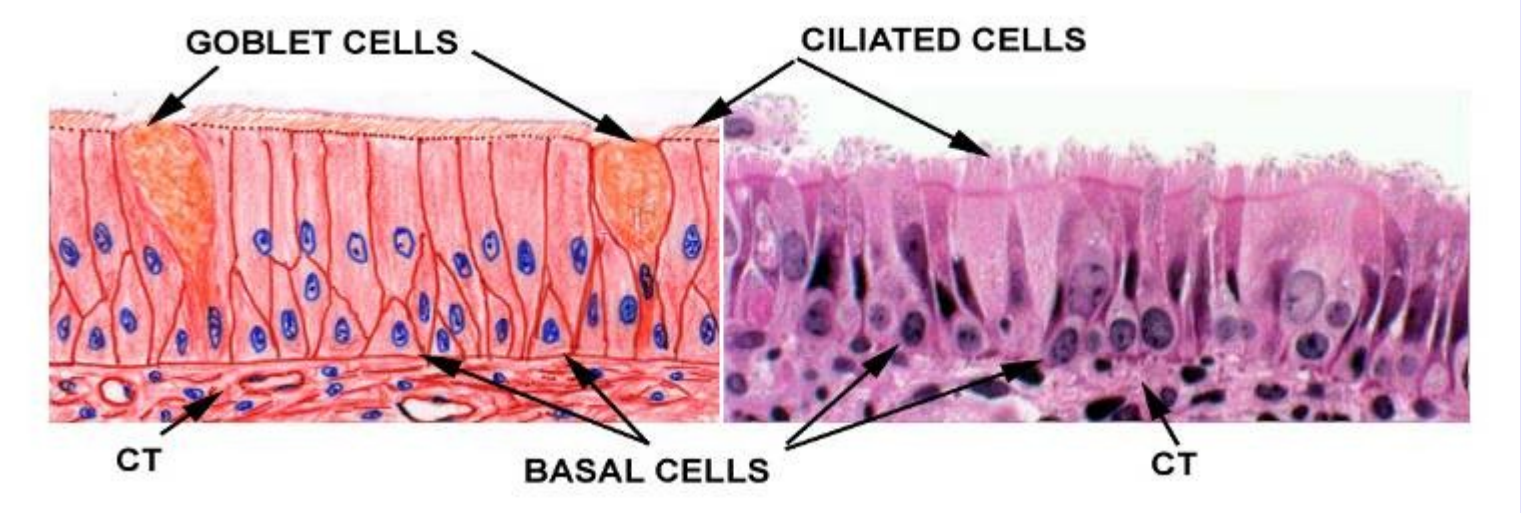
- Second most abundant cell type
- Apical surface packed with membrane bound polysaccharide rich mucous granules
- Function is to secrete mucous to keep surfaces moist and to trap inhaled particles.



Section through trachea showing respiratory epithelium with ciliated cells and goblet cells.



BASAL (SHORT) CELLS



- They are small rounded cells that lie on the basal lamina, but do not extend to the luminal surface of the epithelium
- They are generative cells
- They undergo mitosis and differentiate into other cell types.

SMALL GRANULE CELLS

- Each cell resemble a basal cell, except that it possesses numerous granules (100 – 300 nm in diameter) in the basal cytoplasm
- These endocrine like cells act as effectors in the integration of the mucous and serous secretory processes

CLARA CELLS

- Present in the terminal and respiratory bronchioles.
- They have dome-shaped apices, which protrude into the lumen of the bronchioles.
- They are devoid of cilia.
- They have secretory granules in their apical regions.
- Secrete Glycosaminoglycans, that protect the bronchiolar lining.
- They are also a secondary source of surfactant for the broncheoalveolar fluid



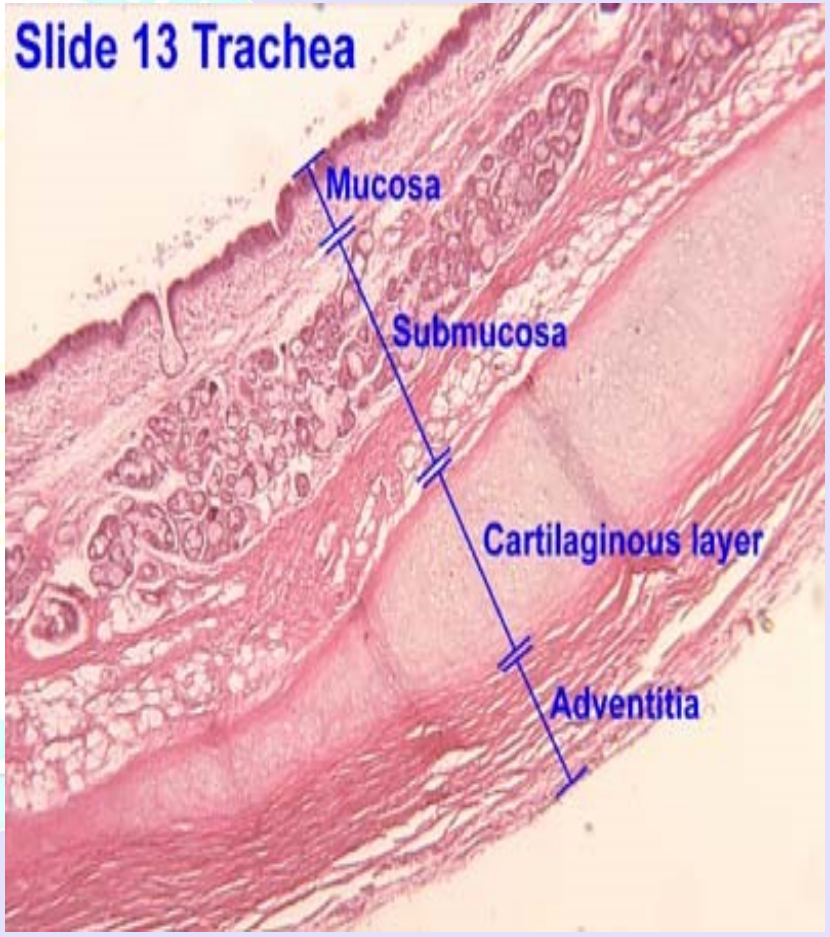
HISTOLOGY OF TRACHEA

TRACHEA

Also called wind pipe is a hollow tube, which transmit the air from the larynx to principal bronchi

As the hollow tube the trachea consists certain layers, these are from out to inside

- Mucosa
- Sub-mucosa
- Adventitia



MUCOSA OF TRACHEA

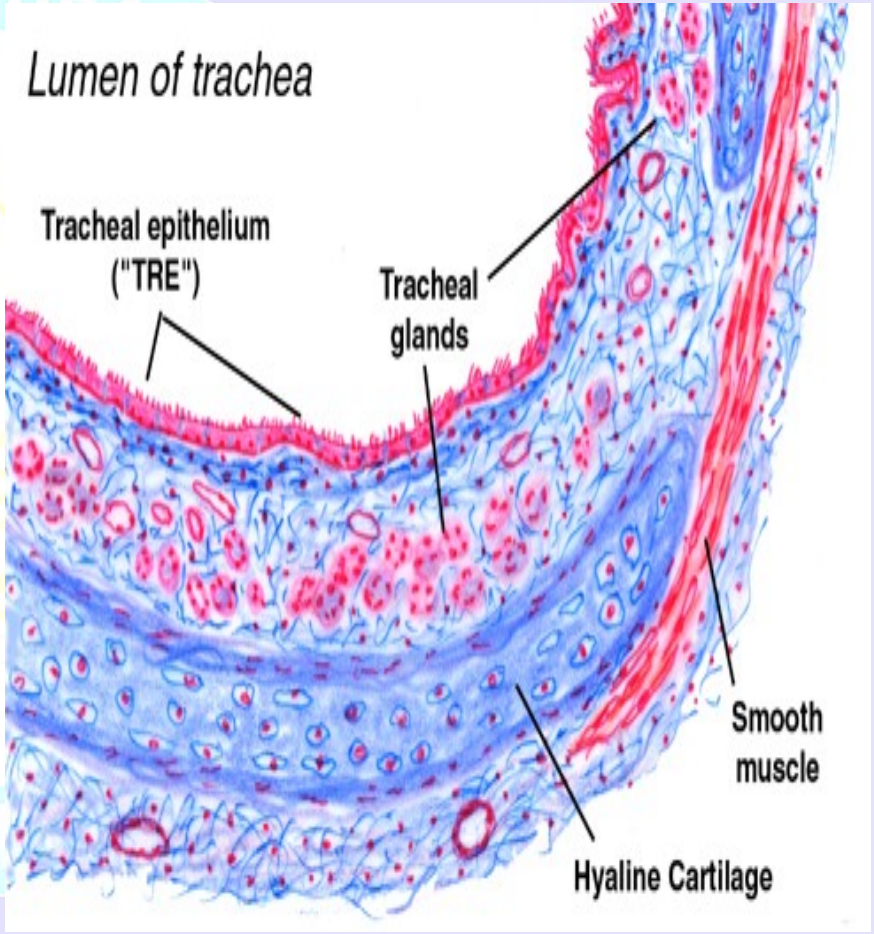
The wall of trachea is formed of following layers from within outwards

MUCOSA:

It is comprised of the following layers.

a. Epithelium of trachea is pseudostratlfled ciliated tall columnar with goblet cells.

The ciliary activity produces constant movements of glandular secretions towards the pharynx. The epithelium is comprised of following cells.



CELLS OF MUCOSA

I. Columnar ciliated cells (each cell bears 300 cilia)

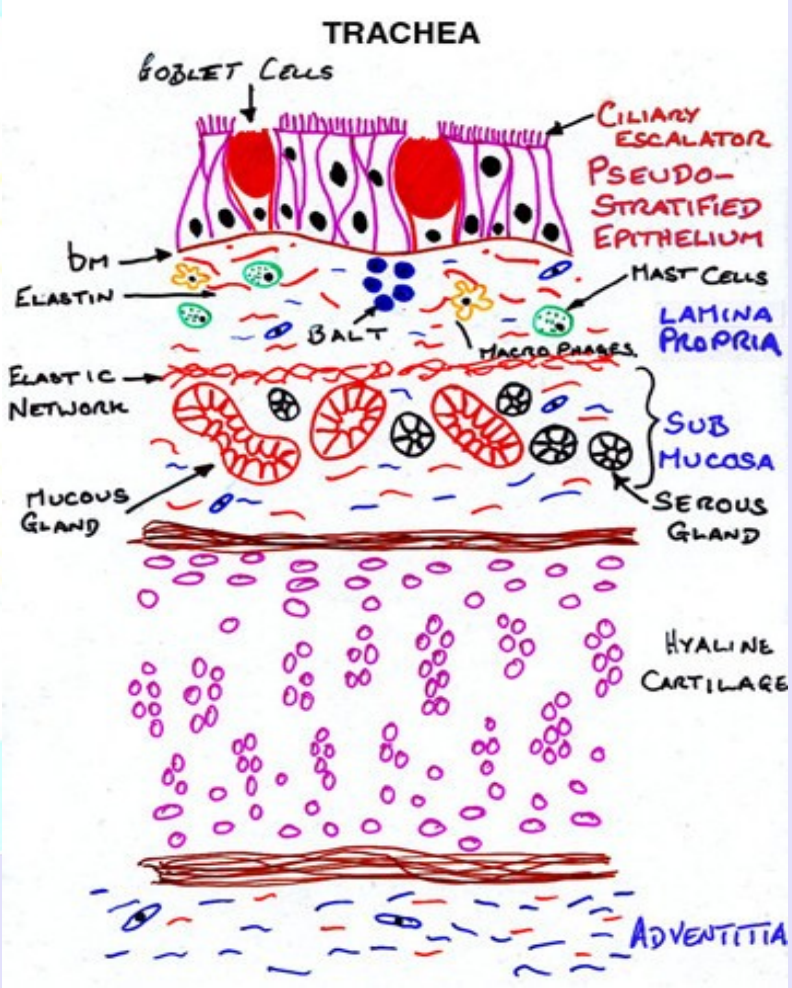
II. Goblet cells (secrete mucous material)

III. Brush cells: These are sensory receptor cells, which are columnar and non- ciliated. Microvilli are present.

IV. Basal or short cells (act as stem cells): These occur in the depth of epithelium, between the bases of the other types of cells and are undifferentiated cells that give rise to goblet cells or ciliated cells.

V. Neuroendocrine cells

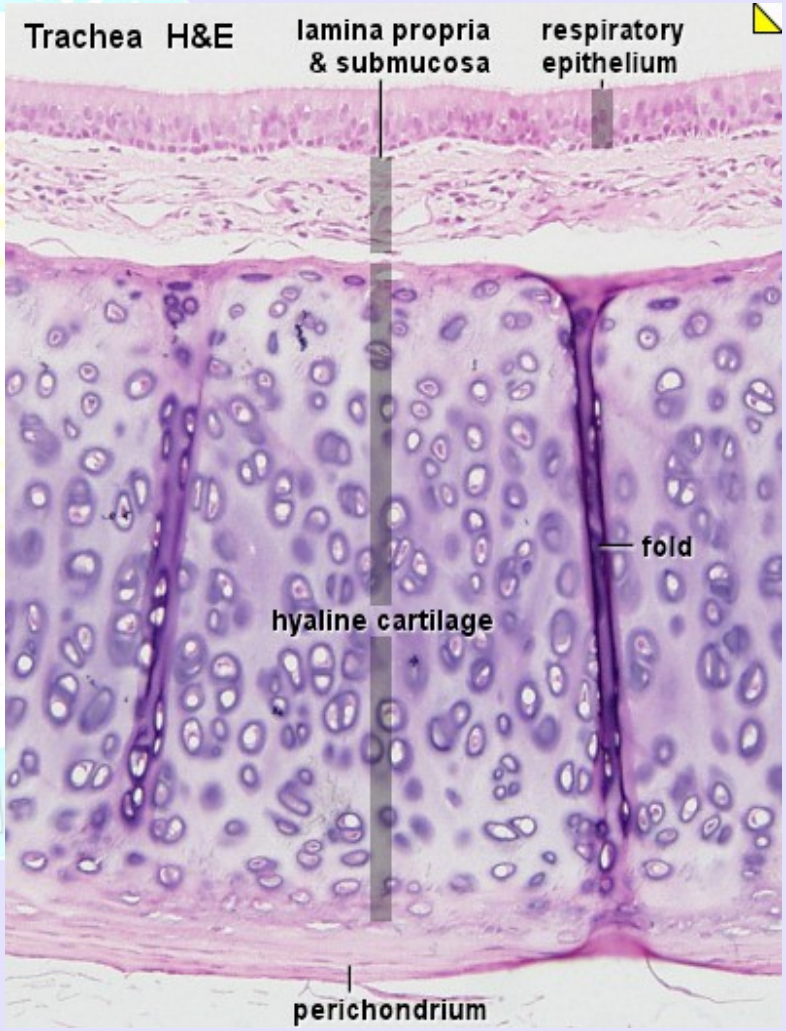
VI. Wandering Cells



Mucosa

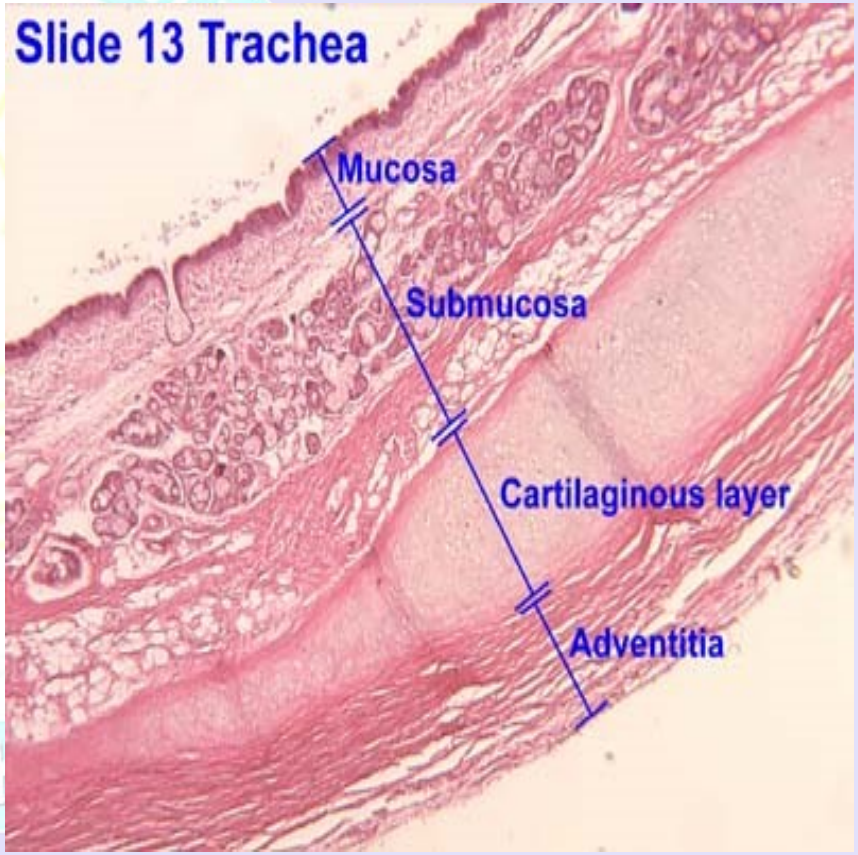
b. The tracheal epithelium is supported by a thick **basement membrane.**

C. Beneath basement membrane **lamina propria** consists of loose highly vascular connective tissue, which becomes condensed at its deeper aspects to form a band of fibroelastic tissue (clear elastic membrane).



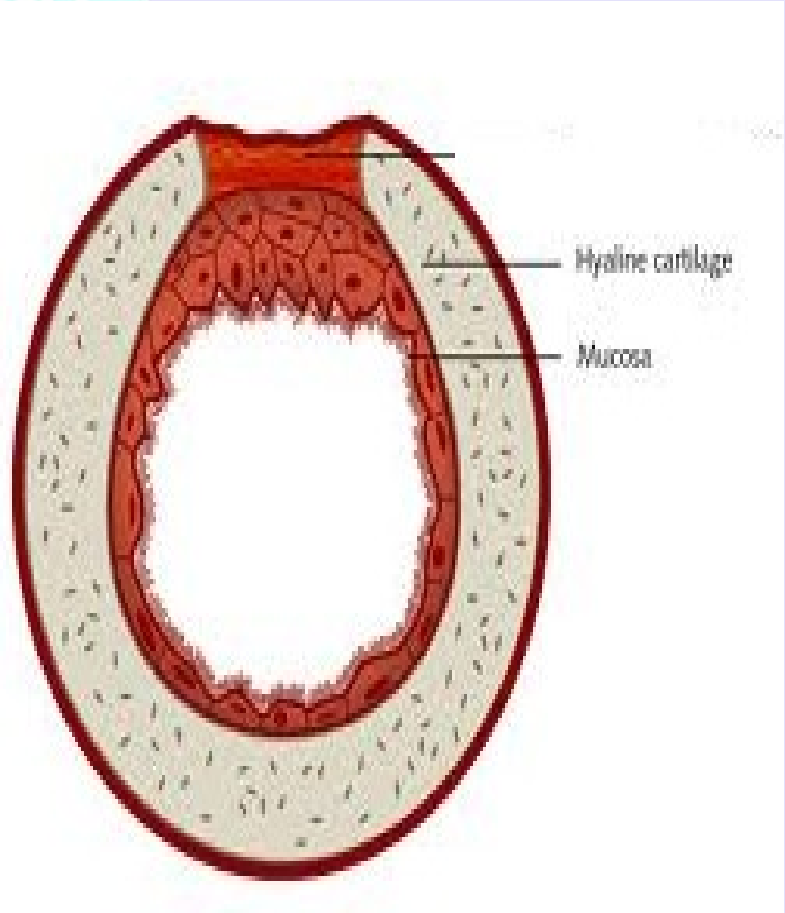
SUBMUCOSA

It contains loose areolar C.T. numerous mixed seromucous and mucous tracheal glands (which decrease in number in lower part of trachea), nerves, vessels and lymphoid follicles.



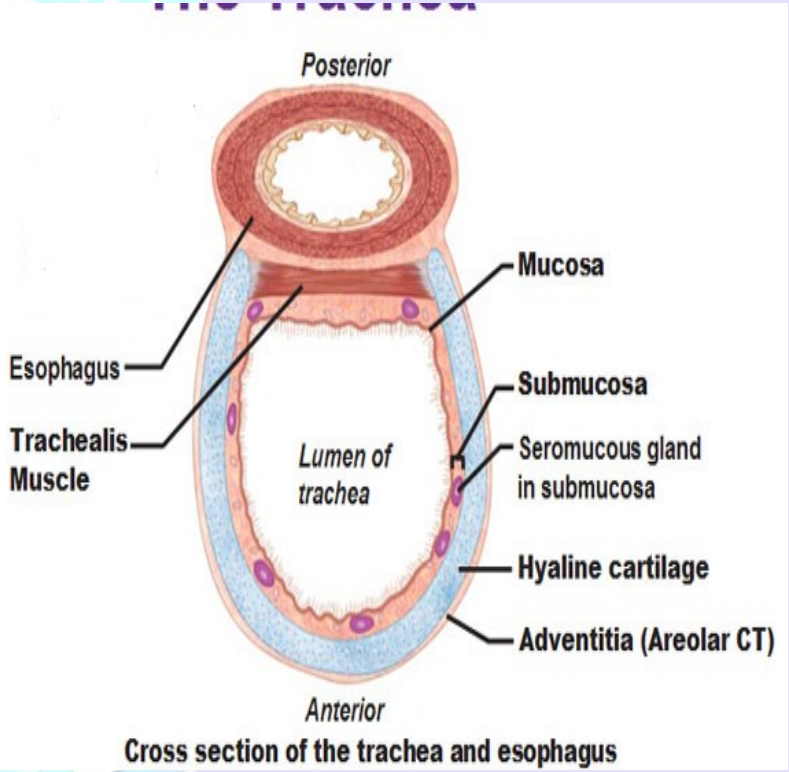
CARTILAGE

- The submucosa merges with the perichondrium of the underlying hyaline cartilage. These cartilages (16-20) are found in the form of C-shaped rings.
- Between the two ends of C-shaped cartilages the submucosa directly merges with the external adventitial layer. Bands of smooth muscles (trachealls) join the free ends of the rings posteriorly. The contraction of muscle and resultant narrowing of the tracheal lumen are useful in cough reflex.



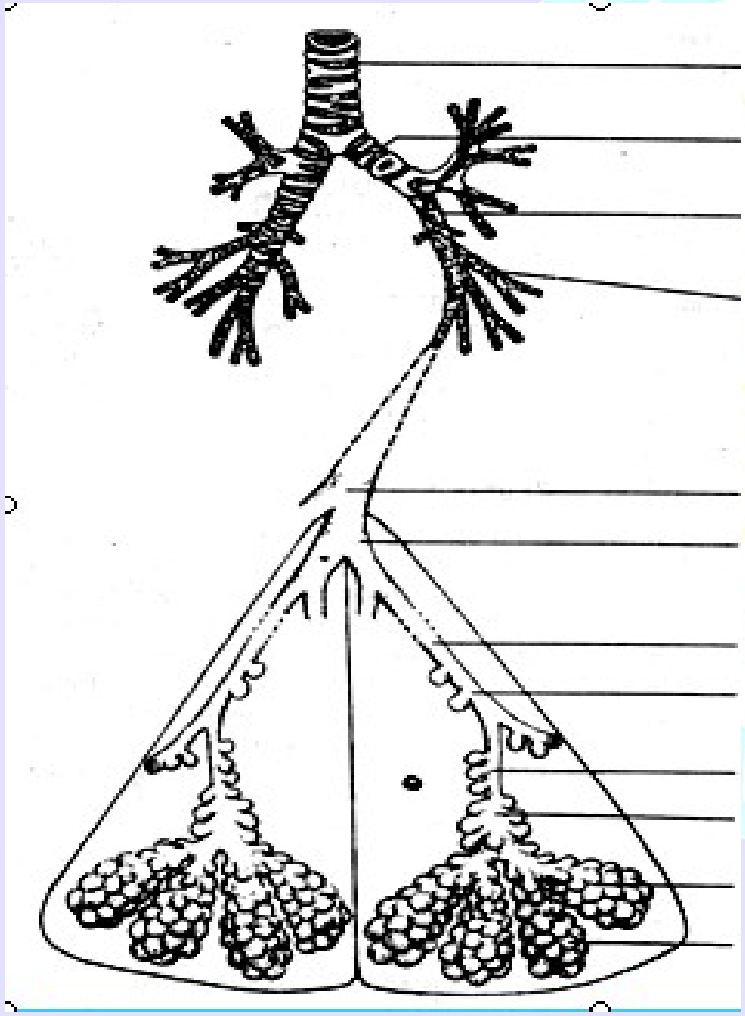
ADVENTITIA /FIBROSA

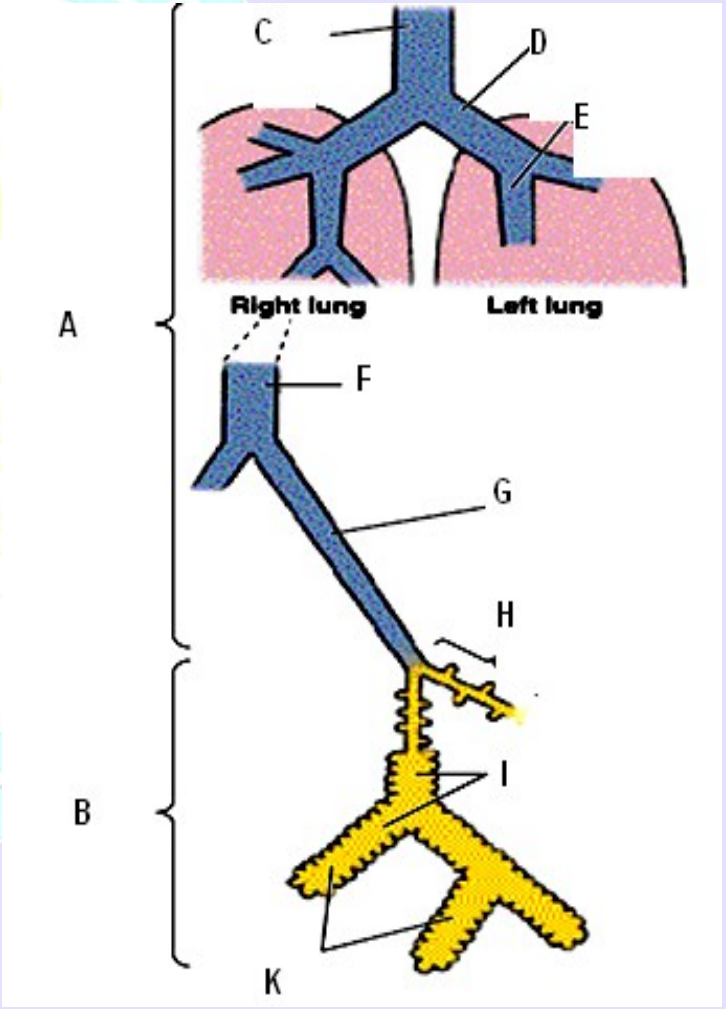
It is formed of areolar C.T. containing small blood vessel and autonomic nerves to trachea.



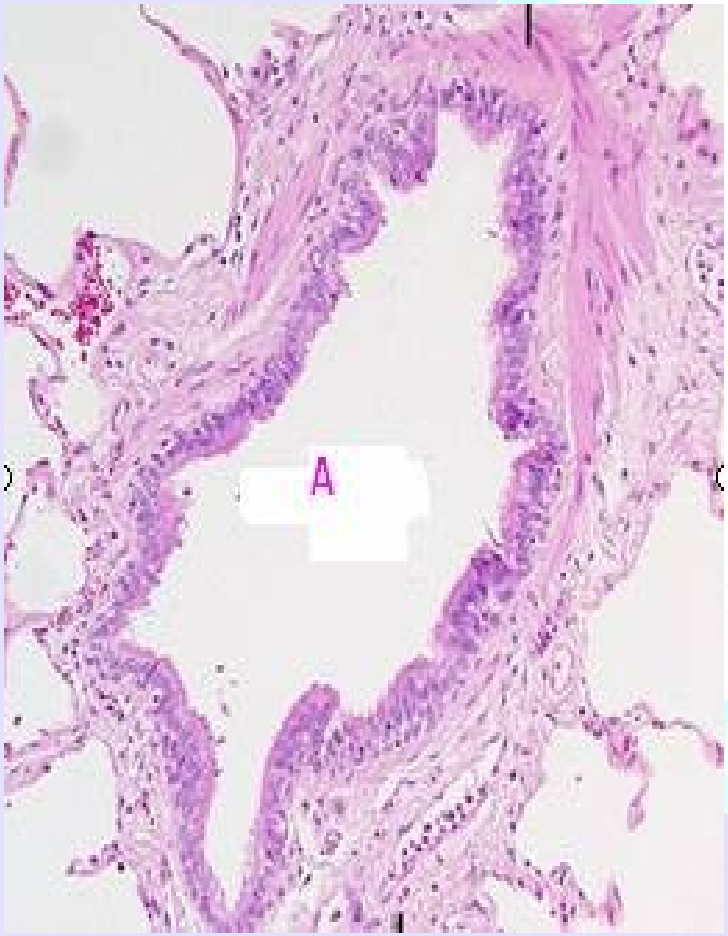
Cross section of the trachea and esophagus

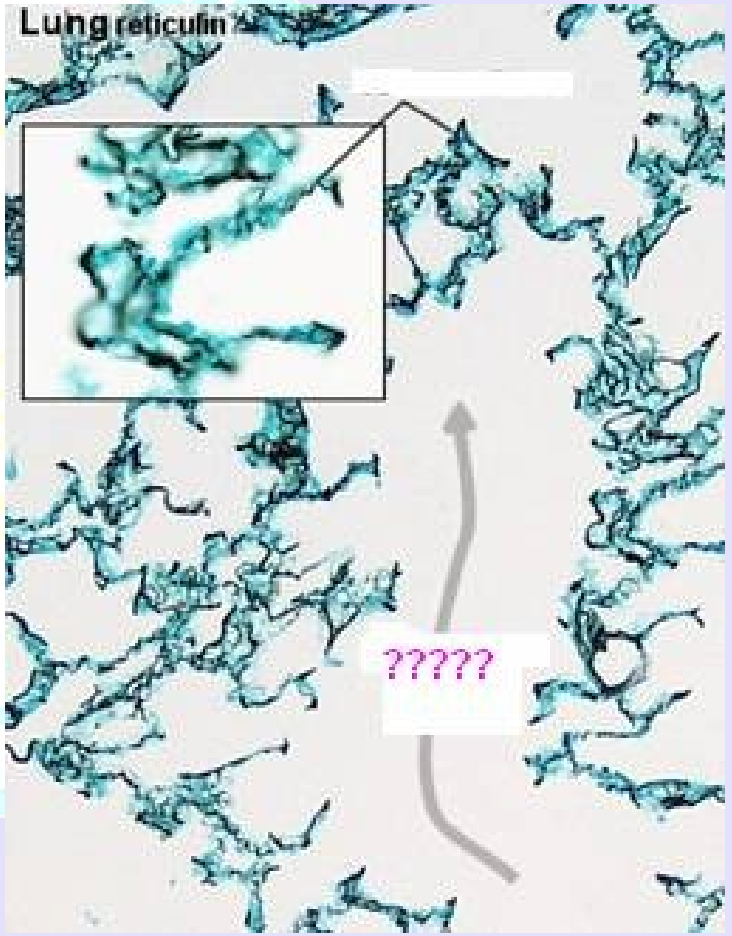
EXERCISE1



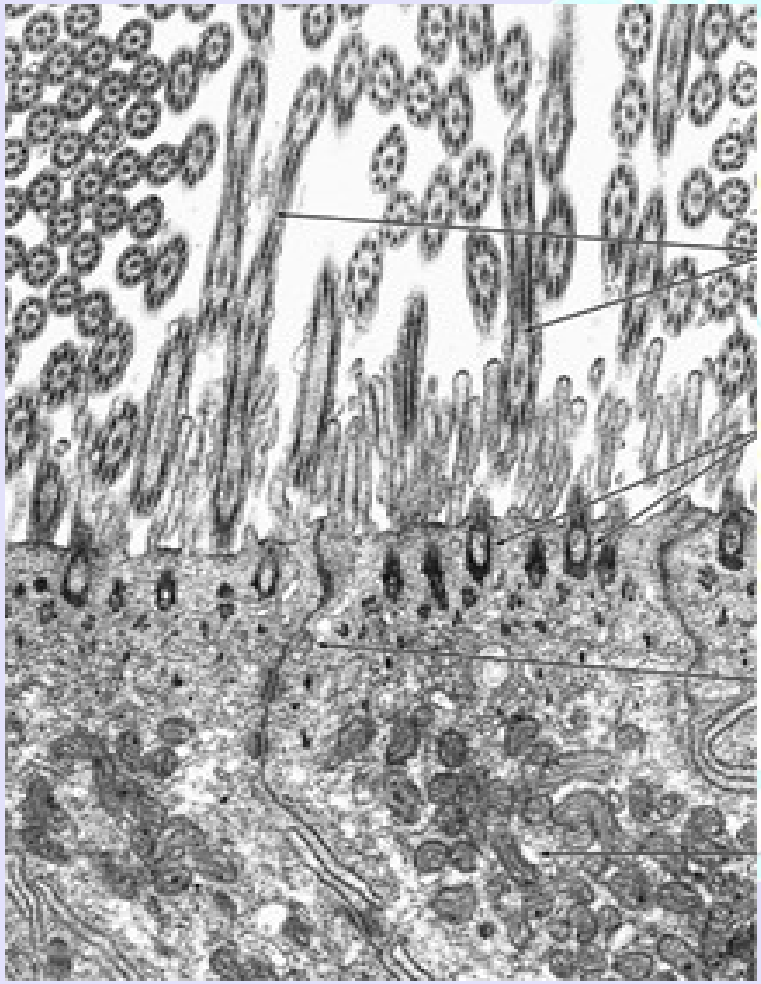


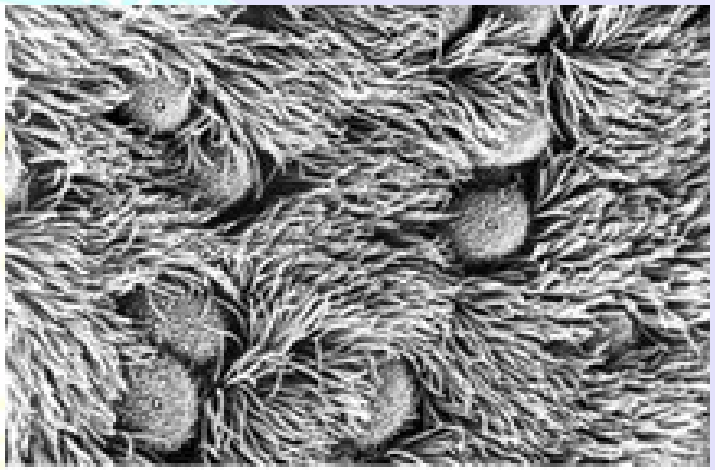
EXERCISE 2

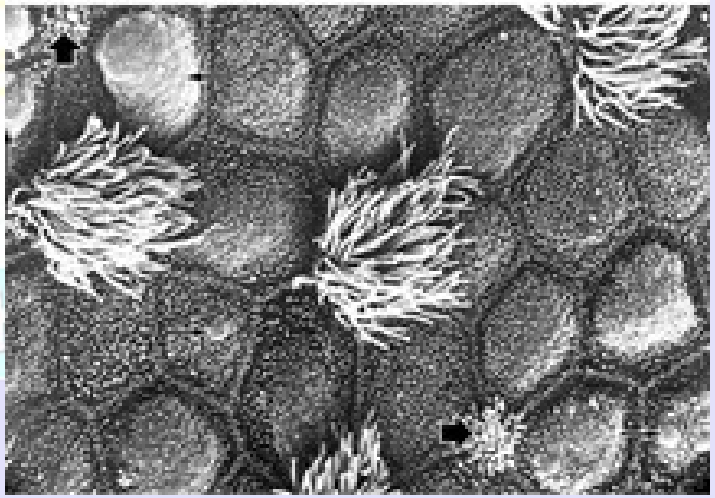




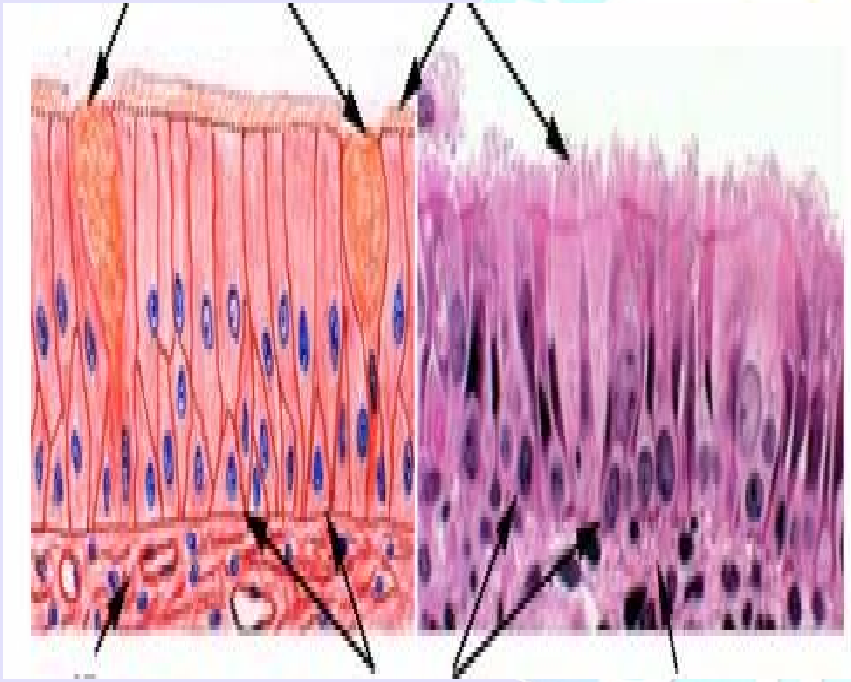
EXERCISE 3

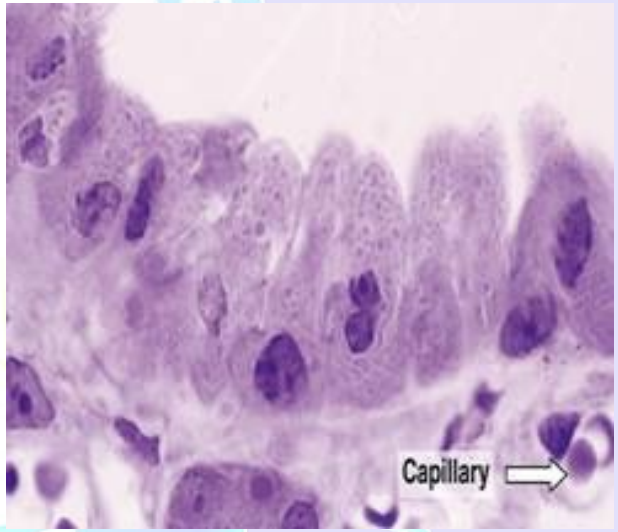




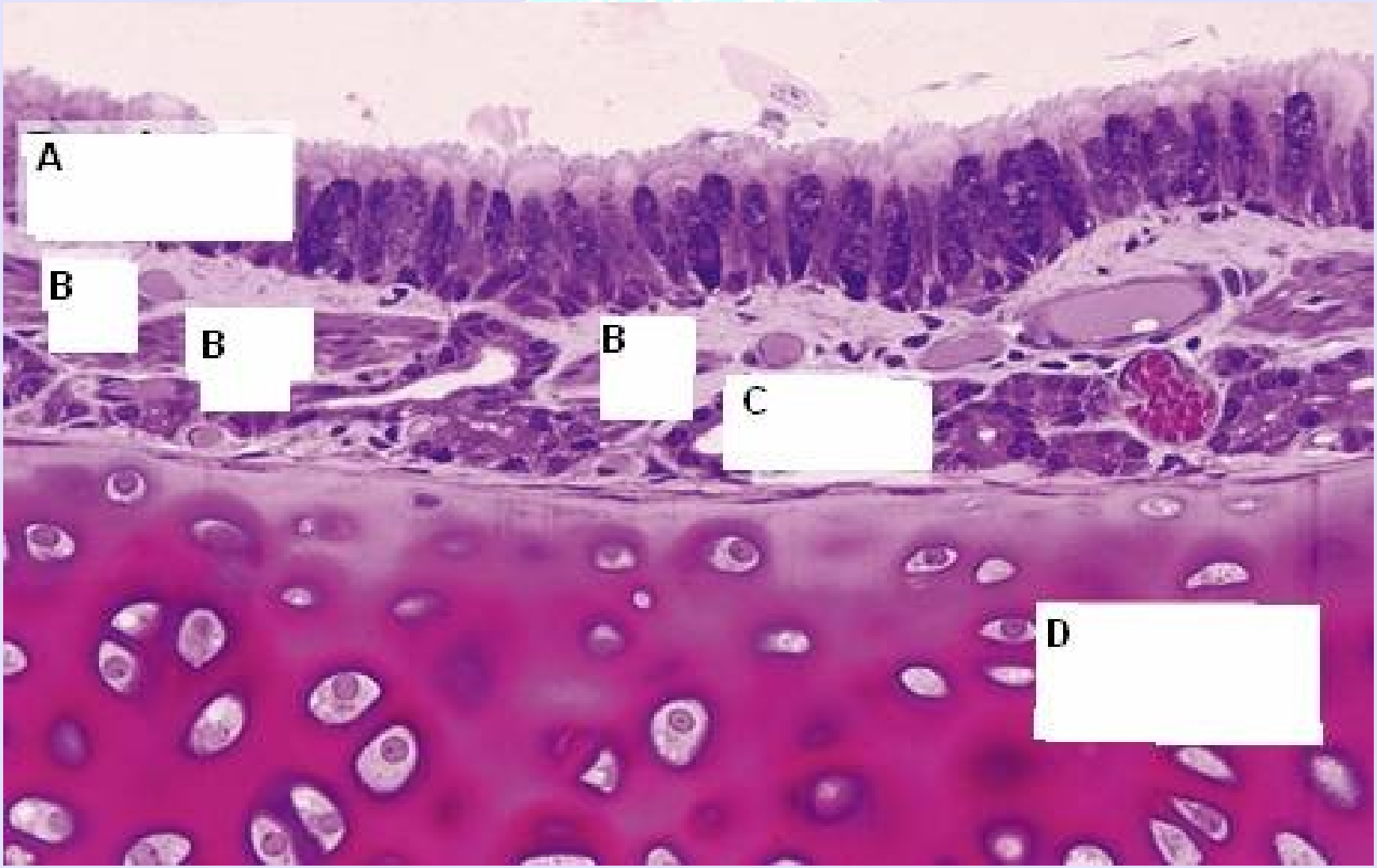


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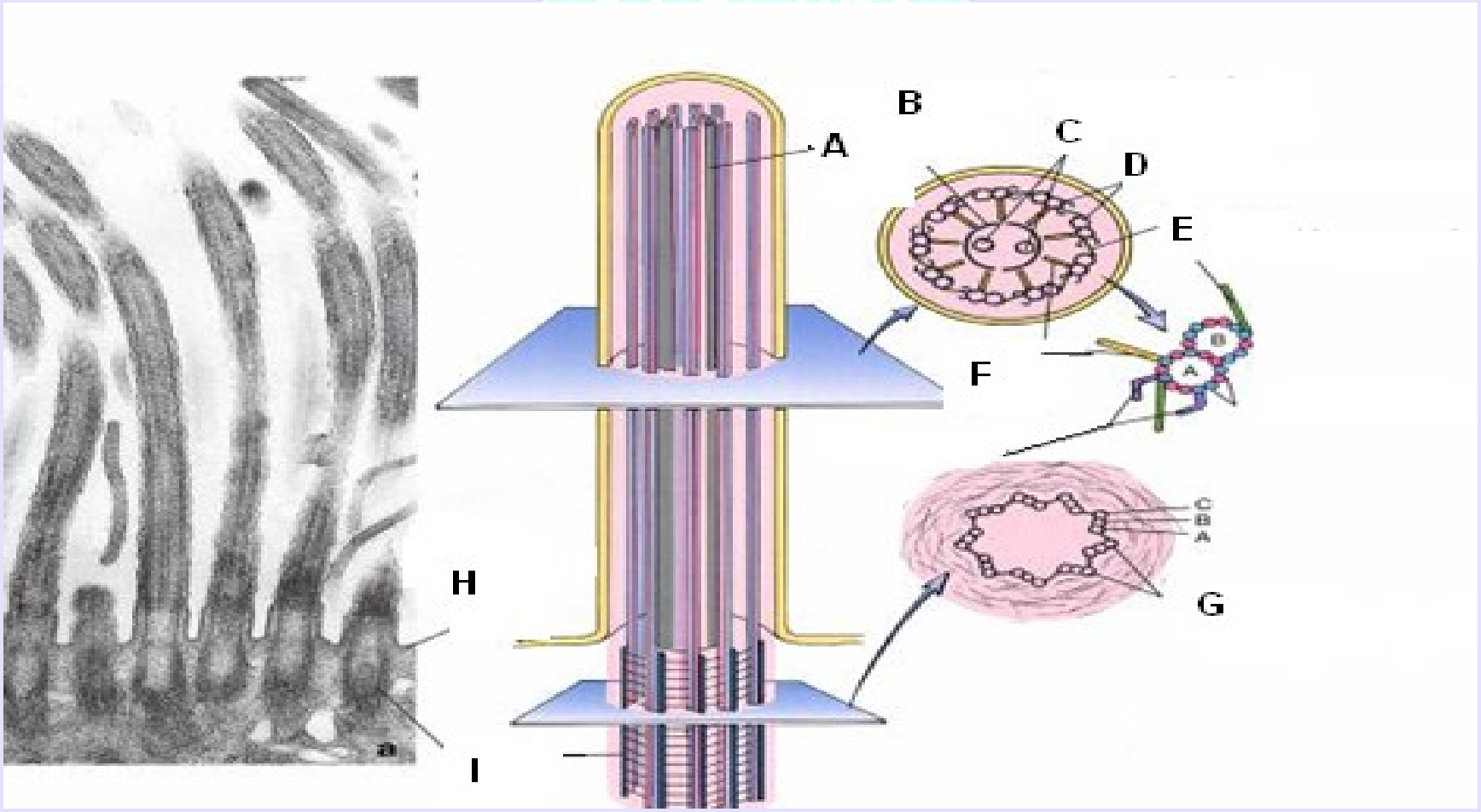




EXERCISE 5



EXERCISE 6



THANK YOU